

Case study practical

1. A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

- Check the charger and power outlet – Ensure the charger and wall socket are working properly.
- Inspect the charging port – Check for damage, loose connections, or charging LED indicators.
- Perform a power reset – Hold the power button for 15–30 seconds after disconnecting power to clear residual charge.
- Test the battery – Remove the battery (if possible) and try powering on with only the charger.
- Check internal hardware – Look for beep codes, fan activity, or indicator lights to identify possible hardware issues.

2. You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.

Hint: Consider checking both hardware connections and basic display settings.

- ⇒ Check monitor power – Ensure the monitor is on.
- ⇒ Check display cable – Secure the HDMI/VGA/DP cable.
- ⇒ Select correct input – Choose the right input source on the monitor.
- ⇒ Restart the PC – Reboot the computer.
- ⇒ Test another cable/monitor – Identify if the cable or monitor is faulty.
- ⇒ Check GPU and display settings – Reseat the GPU and update the graphics driver if needed.

3. A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

- Check the mouse – Ensure it's turned on and replace the battery.
- Test another USB port – Move the USB receiver to a different port.
- Test the mouse on another PC – Check if the mouse itself is faulty.
- Check system settings – Verify the mouse is enabled and drivers are installed.
- Restart the PC – See if the mouse starts working after a reboot.

4. Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.

Hint: Think about fans, dust, and thermal paste.

= i. Dust buildup – Check for dust blocking fans and vents; clean with compressed air.

ii. Faulty cooling fan – Ensure the CPU/GPU fan is spinning properly; replace it if not working.

iii. Dry thermal paste – Inspect the CPU heatsink, clean old thermal paste, and apply fresh thermal paste if needed.